Creative Studio · Nigerian Cultural Intelligence

Add-on architecture and implementation spec for dall-ai (dexway/dall-ai), mapped to the code on `main` as of 31 Oct 2026.

> **The ground rule.** This is purely an addition. No current page, route, table, navigation item, style or flow changes behaviour or appearance. Ads, Marketing, Harvey, the Assistant & launch kit and the existing Content Creator tabs (Brief → Production, Motion video, Autopilot, Usage) stay exactly as they are. Everything new lives in **one new tab group inside Content Creator**, new tables with a `ci_` prefix and a new `lib/culture/` folder.

1. Where it sits in the app today

Existing piece	What it is	How the add-on uses it (read-only unless noted)
`/dashboard/studio` (sidebar label "Content Creator")	Tabbed page; `TABS` array in `page.tsx`	Gets four new tabs appended after "Usage". Existing tabs untouched.
`lib/ai/gateway.ts` `gatewayCall`	The only path to Anthropic: credits, spend caps, web search cost, ledger	New feature names `culture_research`, `culture_curate`, `culture_create`, `culture_qa` added to the union and to `LONG_FEATURES`. Nothing else changes.
`lib/creative/director.ts` research/strategy/calendar/spec	Current Creative Director	Not modified. Approved concepts can be handed to it as a content item (see §6).
`lib/creative/jobs.ts`, `router.ts`, `higgsfield.ts`	Higgsfield generation jobs, budgets, model routing	Production agent creates normal `content_items` / `generation_jobs` through existing functions, so budgets and approvals still apply.
`creative_brands`, `catalogue_items`, `promotions`, `business_profile`, Harvey `knowledge`	Business source of truth	Read for Private Business Memory. Never copied into shared tables.
`lib/net-guard.ts` `fetchPublic`	SSRF-safe fetch	Used by the Source Scout for permitted sources.
`/api/cron/marketing`	Scheduler	Gets one extra call, `runCultureSchedule()`, wrapped in its own try/catch so a failure can't affect existing jobs.
`platform_admins`	Admin gate (only your account)	Gates the new shared-knowledge review page `/admin/culture`.

2. What the user sees

Inside Content Creator, after the existing tabs, a divider labelled **Creative Studio** and four tabs, styled with the existing `studio.css` classes so it looks native:

Tab (`?tab=`)	Purpose
Cultural Research (`culture`)	Ask for research ("funny Lagos relationship situations for a couples' dinner"), pick depth (Quick / Standard / Deep), watch the 7-stage progress panel, read findings with sources and confidence.
Campaign Concepts (`concepts`)	Campaign brief prefilled from the business (type, area, products, offers, colours); objective, creative direction, formats. Returns 3–5 concepts, each with "why this works", the cultural references used and the product tie-in. Approve, revise or reject.
Content Production (`cproduce`)	Approved concept → script, shot list, image/video prompts, carousel, captions, voiceover. QA scorecard. "Send to Production" creates items in the existing Production/Motion pipeline.
Campaign Insights (`insights`)	Enquiries and orders attributed to each campaign (via a tracked WhatsApp link), plus what hooks/formats performed. Shows "not enough data yet" until there is.

No new sidebar item. The existing "Content Creator" label stays. Users never see the agents by name; they see the stages and results.

3. Agent flow (agentic, behind the four tabs)

Agents are **workflow steps sharing the gateway**, each with its own system prompt, allowed tools and output schema (zod, like `lib/creative/schemas.ts`). Not nine always-running models.

A. Cultural Intelligence Orchestrator (shared, runs regardless of any business)

Step	Tools	Writes
Research Director — plans tasks by mode (foundational, scheduled, trend-triggered, on-demand)	none	`ci_research_tasks`
Source Scout + Trend Monitor (one agent in v1)	Anthropic web search; `fetchPublic` on the allow-list	`ci_findings` (status `raw`)
Evidence Verifier	re-fetch, date check, allow-list check	finding status `verified` / `rejected` with reason
Cultural Analyst	retrieval	meaning, audience, region, why it's funny/relatable
Knowledge Archivist	dedupe (normalised key + embedding similarity later)	`ci_knowledge` new version, status `pending_review`

Shared knowledge becomes `trusted` only after admin approval at `/admin/culture` (or automatic approval for high-confidence evergreen items, behind a setting that defaults off). Research updates the knowledge base, never the model, so every fact can be traced, corrected and rolled back.

B. Commercial Creative Orchestrator (per business, per campaign)

```
Brief → Brand Strategist → Cultural Analyst (retrieve trusted + campaign research)
      → [if depth ≠ Quick: on-demand research task, results shown as they land]
      → Story Architect (3–5 concepts) → user approves one
```

```
      → Production Director (package) → QA (scorecard; one auto-fix retry)
      → user approves → existing content_items / generation_jobs
```

QA reuses the pattern of Harvey's `gate.ts`: deterministic checks (claims match catalogue prices and active promotions, no banned phrases, no unfilled placeholders) plus an AI review of language, cultural fit and freshness. A failed check shows the reason, as Harvey does.

C. Learning loop

Performance Analyst (phase 7) reads campaign attribution and reports per audience and format, with sample sizes. It writes to `ci_performance`, never directly to shared knowledge.

4. Data model (new tables only)

Migration `0028_creative_studio_culture.sql`. Shared tables have no `workspace_id` and are readable by signed-in users, writable only by the service role. Tenant tables use the existing `is_member(workspace_id)` RLS.

Table	Memory	Key columns
`ci_sources`	shared	url/domain, kind (blog, forum, news, comedy, api), allowed bool, terms_note, added_by
`ci_research_tasks`	shared or campaign	mode, topic, depth, workspace_id (null = shared), campaign_id, status, stage, error, cost_usd
`ci_findings`	evidence	task_id, source_url, published_at, discovered_at, excerpt (short, no copying), status, reject_reason
`ci_knowledge`	shared cultural	key, version, title, meaning, kind (slang, situation, reference, trend, history), lifespan (evergreen, recurring, trending, historical), region, audience, evidence ids, confidence, usage_rights, status, last_verified_at, refresh_due_at, superseded_by
`ci_campaigns`	campaign	workspace_id, brief jsonb, depth, status, chosen_concept_id, tracking_code
`ci_concepts`	campaign	campaign_id, title, story, why, knowledge ids, product refs, status, feedback
`ci_packages`	campaign	concept_id, script, shots, prompts, captions, qa jsonb, content_item_ids
`ci_performance`	performance	campaign_id, period, views, clicks, enquiries, orders, revenue_ngn, source

Private business memory is not a new table: it is assembled at runtime from the existing business tables, the same way Harvey's `gatherAccount` does, so no business data is ever written into shared rows.

5. Research freshness and sources

Lifespan	Default re-verify
Evergreen language / culture	Quarterly
Historical references	When new evidence appears

Recurring social situations	Monthly
Lifestyle / entertainment trends	Weekly
Live viral topics	Daily, permitted sources only
Campaign research	On demand

> **Source rules.** No scraping of Instagram, Facebook or TikTok; their terms forbid unapproved automated collection. Use Anthropic web search, an admin-managed allow-list of blogs, news and forums whose terms permit it, official APIs when connected, and human-curated entries. Store short excerpts and summaries, never whole articles or posts, and always the source and date.

6. Hand-off into the existing pipeline

- "Send to Production" creates `content_items` rows in the brand's current calendar month (or a new "Creative Studio" calendar) with the package's prompt. From there the existing Production tab, budgets, model router and Higgsfield jobs do the work unchanged.
- Launch-style concepts can prefill a `motion_projects` brief instead.
- Each campaign gets a `tracking_code`; its WhatsApp link is `wa.me/<number>?text=…[code]`. The inbox already records the first message, so attribution is one new lookup in the existing conversation start handler, read-only for everything else.
- Ads stays separate. A campaign can later be offered to Ads as a creative, but nothing in Ads changes in this work.

6b. Shared cultural context for the other agents (read-only)

Once trusted knowledge exists, one small helper, `cultureContext(workspaceId, topic, max)` in `lib/culture/retrieve.ts`, returns a short block (a few hundred tokens) of relevant trusted references for that business's niche and area. Other agents can opt in by adding that block to their prompt; they keep their own logic, UI and outputs:

Agent	What it gains	Touch point
Harvey (writer, handler)	Outreach and replies that sound local and current, with references a Lagos owner recognises	One line appended in `knowledgeBlock()`; Harvey's gate still checks every draft
Motion video (`MOTION_SKILL`)	Launch-video scenes, captions and on-screen lines grounded in recognisable situations	Extra context in `motionRequest()`
Sales assistant (`lib/ai/agent.ts`)	Tone and expressions only; prices and offers still come only from the catalogue	Optional, off by default, per-agent toggle
Creative Director research / calendar	Content ideas tied to current cultural moments	Context block in `runResearch` / `draftCalendar`

Each hook is guarded: if there's no trusted knowledge or the lookup fails, the agent runs exactly as it does today. These come after phase 2, one agent at a time.

7. Files to add (and the only lines touched in existing files)

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NEW  lib/culture/agents.ts        prompts + zod schemas for each agent
NEW  lib/culture/orchestrator.ts  runCultureTask, runCampaign, stage updates
NEW  lib/culture/retrieve.ts      controlled retrieval across the 4 memories
NEW  lib/culture/qa.ts            deterministic checks + AI review
NEW  lib/culture/schedule.ts      runCultureSchedule (freshness, trend checks)
NEW  app/dashboard/studio/culture/*.tsx   the four tab panels + actions.ts
NEW  app/admin/culture/page.tsx   review queue, source allow-list (platform_admins only)
NEW  supabase/migrations/0028_creative_studio_culture.sql
NEW  tests/culture.test.ts

EDIT lib/ai/gateway.ts            +4 feature names (type + LONG_FEATURES)
EDIT app/dashboard/studio/page.tsx  append 4 entries to TABS, 4 render lines
EDIT app/api/cron/marketing/route.ts  +1 guarded call
```

8. Build order

Phase	Delivers	Needs
1	Migration, orchestrator skeleton, stage tracking, Cultural Research tab (on-demand), admin review	Anthropic credit
2	Foundational research seed (Lagos life, comedy, slang, history), scheduled refresh	Allow-list of sources
3	Campaign Concepts tab: strategist + analyst + Story Architect	—
4	Content Production tab: package + QA + hand-off to Production/Motion	Higgsfield models enabled
5	Tracking links + Campaign Insights	WhatsApp channel connected
6	Performance learning feeding concept ranking	Real campaign data

9. Cost and safety controls

- All calls go through `gatewayCall`, so existing credits, spend caps and ledgers apply. Deep research counts its web searches.
- Each workspace gets a daily cap on research tasks; shared scheduled research has its own monthly USD budget set in `/admin/culture`.
- Nothing publishes automatically. Every concept and package needs the owner's approval, as in the current Content Creator.
- Tenant isolation: retrieval takes the workspace id and only ever joins shared + that workspace's rows; tests cover it.

10. Open decisions for you

- Should shared knowledge go live only after your review, or auto-approve high-confidence evergreen items?

- Which sources to start the allow-list with (e.g. Nigerian news sites, Nairaland sections whose terms allow it, comedy blogs)?
- Monthly budget for shared background research.